In [4]:
```
import sqlite3
import pandas as pd
import numpy as np
```

In [20]:
```
sql = sqlite3.connect(r"C:\Users\lenovo\Downloads\SQLite Test.db")
df=pd.read_sql_query("SELECT * FROM dataset_1", sql)
```

In [21]:
```
# SELECT * FROM dataset_1
df
```

Out[21]:

	destination	passanger	weather	temperature	time	coupon	expiratior
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1c
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1c
...	...	...	...	...	...	...	..
12679	Home	Partner	Rainy	55	6PM	Carry out & Take away	1c
12680	Work	Alone	Rainy	55	7AM	Carry out & Take away	1c
12681	Work	Alone	Snowy	30	7AM	Coffee House	1c
12682	Work	Alone	Snowy	30	7AM	Bar	1c
12683	Work	Alone	Sunny	80	7AM	Restaurant(20-50)	2h

12684 rows × 27 columns

In [18]:
```
# SELECT * FROM dataset_1 limit 10
df.head(10)
```

Out[18]:

	destination	passanger	weather	temperature	time	coupon	expiration	ge
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	Fe
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	Fe
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	Fe
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	Fe
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	Fe
5	No Urgent Place	Friend(s)	Sunny	80	6PM	Restaurant(<20)	2h	Fe
6	No Urgent Place	Friend(s)	Sunny	55	2PM	Carry out & Take away	1d	Fe
7	No Urgent Place	Kid(s)	Sunny	80	10AM	Restaurant(<20)	2h	Fe
8	No Urgent Place	Kid(s)	Sunny	80	10AM	Carry out & Take away	2h	Fe
9	No Urgent Place	Kid(s)	Sunny	80	10AM	Bar	1d	Fe

10 rows × 27 columns

In [15]:

```
pd.read_sql_query("SELECT * FROM dataset_1 LIMIT 10",sql)
```

Out[15]:

	destination	passanger	weather	temperature	time	coupon	expiration	ge
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	Fe
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	Fe
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	Fe
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	Fe
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	Fe
5	No Urgent Place	Friend(s)	Sunny	80	6PM	Restaurant(<20)	2h	Fe
6	No Urgent Place	Friend(s)	Sunny	55	2PM	Carry out & Take away	1d	Fe
7	No Urgent Place	Kid(s)	Sunny	80	10AM	Restaurant(<20)	2h	Fe
8	No Urgent Place	Kid(s)	Sunny	80	10AM	Carry out & Take away	2h	Fe
9	No Urgent Place	Kid(s)	Sunny	80	10AM	Bar	1d	Fe

10 rows × 27 columns

In [23]:
```
df
```

Out[23]:

	destination	passanger	weather	temperature	time	coupon	expiratior
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1c
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1c
...	...	...	...	...	...	...	..
12679	Home	Partner	Rainy	55	6PM	Carry out & Take away	1c
12680	Work	Alone	Rainy	55	7AM	Carry out & Take away	1c
12681	Work	Alone	Snowy	30	7AM	Coffee House	1c
12682	Work	Alone	Snowy	30	7AM	Bar	1c
12683	Work	Alone	Sunny	80	7AM	Restaurant(20-50)	2h

12684 rows × 27 columns

In [26]:
```
df["passanger"].unique()
```

Out[26]:
```
array(['Alone', 'Friend(s)', 'Kid(s)', 'Partner'], dtype=object)
```

In [27]:
```
df[df['destination']=='Home']
```

Out[27]:

	destination	passanger	weather	temperature	time	coupon	expiration
13	Home	Alone	Sunny	55	6PM	Bar	1c
14	Home	Alone	Sunny	55	6PM	Restaurant(20-50)	1c
15	Home	Alone	Sunny	80	6PM	Coffee House	2h
35	Home	Alone	Sunny	55	6PM	Bar	1c
36	Home	Alone	Sunny	55	6PM	Restaurant(20-50)	1c
...	...	...	...	...	...	...	..
12675	Home	Alone	Snowy	30	10PM	Coffee House	2h
12676	Home	Alone	Sunny	80	6PM	Restaurant(20-50)	1c
12677	Home	Partner	Sunny	30	6PM	Restaurant(<20)	1c
12678	Home	Partner	Sunny	30	10PM	Restaurant(<20)	2h
12679	Home	Partner	Rainy	55	6PM	Carry out & Take away	1c

3237 rows × 27 columns

In [28]:
```
df.sort_values('coupon')
```

Out[28]:

	destination	passanger	weather	temperature	time	coupon	expiration
11702	Home	Partner	Sunny	30	10PM	Bar	2h
9930	No Urgent Place	Alone	Snowy	30	2PM	Bar	1d
10632	Home	Alone	Rainy	55	6PM	Bar	1d
7997	No Urgent Place	Friend(s)	Rainy	55	10PM	Bar	2h
11166	Work	Alone	Snowy	30	7AM	Bar	1d
...	...	...	...	...	...	...	..
10476	Home	Alone	Sunny	80	6PM	Restaurant(<20)	1d
5447	Home	Alone	Sunny	80	10PM	Restaurant(<20)	2h
10478	Home	Alone	Snowy	30	10PM	Restaurant(<20)	2h
5440	No Urgent Place	Alone	Sunny	80	2PM	Restaurant(<20)	2h
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d

12684 rows × 27 columns

In [30]:
```
df.rename(columns={'destination':'Destination'},inplace=True)
```

In [31]:
```
df
```

Out[31]:

	Destination	passanger	weather	temperature	time	coupon	expiratio
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1
...	...	...	...	...	...	...	.
12679	Home	Partner	Rainy	55	6PM	Carry out & Take away	1
12680	Work	Alone	Rainy	55	7AM	Carry out & Take away	1
12681	Work	Alone	Snowy	30	7AM	Coffee House	1
12682	Work	Alone	Snowy	30	7AM	Bar	1
12683	Work	Alone	Sunny	80	7AM	Restaurant(20-50)	2

12684 rows × 27 columns

In [32]:
```
df.groupby('occupation').size().to_frame('Count').reset_index()
```

Out[32]:

	occupation	Count
0	Architecture & Engineering	175
1	Arts Design Entertainment Sports & Media	629
2	Building & Grounds Cleaning & Maintenance	44
3	Business & Financial	544
4	Community & Social Services	241
5	Computer & Mathematical	1408
6	Construction & Extraction	154
7	Education&Training&Library	943
8	Farming Fishing & Forestry	43
9	Food Preparation & Serving Related	298
10	Healthcare Practitioners & Technical	244
11	Healthcare Support	242
12	Installation Maintenance & Repair	133
13	Legal	219
14	Life Physical Social Science	170
15	Management	838
16	Office & Administrative Support	639
17	Personal Care & Service	175
18	Production Occupations	110
19	Protective Service	175
20	Retired	495
21	Sales & Related	1093
22	Student	1584
23	Transportation & Material Moving	218
24	Unemployed	1870

In [33]:
```
df.groupby('weather')['temperature'].mean().to_frame('avg_temp').reset_index()
```

Out[33]:

	weather	avg_temp
0	Rainy	55.000000
1	Snowy	30.000000
2	Sunny	68.946271

In [34]:
```
df.groupby('weather')['temperature'].size().to_frame('Count_temp').reset_index()
```

Out[34]:

	weather	Count_temp
0	Rainy	1210
1	Snowy	1405
2	Sunny	10069

In [35]:
```
df.groupby('weather')['temperature'].nunique().to_frame('count_distinct_temp').r
```

Out[35]:

	weather	count_distinct_temp
0	Rainy	1
1	Snowy	1
2	Sunny	3

In [36]:
```
df.groupby('weather')['temperature'].sum().to_frame('sum_temp').reset_index()
```

Out[36]:

	weather	sum_temp
0	Rainy	66550
1	Snowy	42150
2	Sunny	694220

In [ ]: